

Internal Combustion Engine exercise

Start from the template model in the path:

...\GTI\v2016\examples\Engine_1D_Gas_Exchange_Combustion\Gasoline\SI_4cyl_SITurb_PortFuelFilm_TW allSoln.

and scale the bore and stroke of the cylinder according to the percentage assigned individually. Scale the intake and exhaust valve diameter according to the bore variation. Consider a fixed air-to-fuel ratio equal to the stoichiometric value of gasoline.

Investigate the effects of the variations of:

1. engine speed (1000, 2000, 3000, 4000, 5000 rpm);
2. repeat the engine speed sweep with variations of the air-to-fuel ratio (+ 20% and – 20% with respect to the initial value);
3. spark advance for a fixed engine speed, applying variations of -10, -5, +5, +10 crank angle degrees;
4. throttle valve position (use throttle valve controller taken from “..\GTI\v2016\examples\Engine_1D_Gas_Exchange_Combustion\Gasoline\SI_4cyl_ThrottleController”) (50%, 20%, 10%, and 5% of maximum torque) for a fixed engine speed;

looking to:

- BMEP, IMEP, brake torque against engine speed (points 1,2,3);
- Brake power against engine speed (points 1,2,3);
- Volumetric efficiency against engine speed (points 1,2,3);
- Brake specific fuel consumption against engine speed (points 1,2,3) or against the load (point 4);
- Pressure cycle in p-V and p-theta diagrams.

Justify the effects based on mathematical relationships among the above parameters.

Vehicle simulation

Simulate PX vehicle along WLTC class3 cycles controlled by ECMS, using constant equivalence factor (EF) and adapting the maximum torque of the ICE according to the individual assignment. Tune the EF to have the SOC balance. Run the simulation with variable “adaptive” EF.

Without modifying ECMS settings, run analyses with a different driving cycle, selected among the following options:

- FTP75
- LA92
- UDDS
- Artemis Rural Road
- Artemis Motorway 130 kmhr
- JC08 Hot
- Braunschweig City driving cycle
- Heavy Heavy-duty Diesel Truck (HHDDT) – Cruise Mode
- JE05
- CUEDC diesel cycle – NB
- China Light-Duty Vehicle Test Cycle for Passenger Car

Extract instantaneous trends of:

- SOC
- ICE and EM power, rotational speed and torque;
- Emissions of CO₂, CO, NO_x and HC (units g/km);
- Fuel economy (units l/100 km).

Compare emissions with gasoline EURO 6e and EU CO₂ target @ 2025. Extract and comment instantaneous operating points of the ICE and EM over BSFC and efficiency maps, respectively.

Without modifying ECMS settings, repeat previous steps applying a variation of + and – 25% to at least one of the following data:

- Vehicle mass
- Vehicle frontal area
- Battery maximum capacity

Compare all the previous parameters and in particular the instantaneous operating points of ICE and EM over BSFC and efficiency maps, respectively, for the different vehicle configurations and justify the variations based on theoretical basis.